

Project Overview

Fleet vehicles used for specialised operations such as emergency response, firefighting, forestry, utilities, and other off-road or service fleets operate in demanding environments and duty cycles that differ significantly from production passenger vehicles. These vehicles often experience repeated high inrush and extended cranking events, with additional auxiliary loads driven by mission-specific equipment.

In many cases, OEM electrical systems are intentionally designed with limited protective devices around starting and power distribution circuits to avoid nuisance trips or impaired availability during critical operations. However, fleet operators and clients increasingly require an additional level of electrical protection to reduce the risk of cable overheating, insulation damage, or thermal incidents, particularly in modified or heavily loaded systems.

This project focuses on improving electrical protection strategies for specialised fleet vehicles by developing and enhancing a tool that calculates and validates suitable cable and fuse combinations. The objective is to achieve a balanced level of protection mitigating thermal and fault risks while preserving reliable performance under high inrush and cranking conditions.

We are seeking a motivated intern/s to support the next phase or refine the development of this engineering tool, currently implemented in Excel and aimed at having a mobile-friendly application for field use.

Key Responsibilities

- Validate and expand existing electrical models and calculations, including:
 - Cable sizing
 - Fuse selection under high-current and engine cranking scenarios
 - Voltage drop analysis and thermal limit assessment
 - Integrate manufacturer datasheets and relevant industry standards (e.g. IEC, AS/NZS, SAE) into the tool's component and standards libraries
- Work closely with the engineering team to:
 - Streamline and improve the underlying Excel logic
 - Support migration of the tool to app-based platforms (e.g. MATLAB App Designer or Python-based GUIs)
- Document:
 - Use cases
 - Logic flow
 - Key engineering assumptions to support verification, validation, and maintainability
- Assist with commissioning the tool for trial deployment and update it based on feedback from field use
- Participate in technical reviews, testing, and deployment of the updated tool
- Develop and maintain a database of fleet assets (vehicles) and associated electrical components

- Review real-world case studies and field data to identify root causes of electrical and thermal issues
- *(Optional)* Conduct benchmarking against other industry tools or available field data

Project Phase Breakdown

Phase 1: Develop the User Requirements

- Work with the engineering team to define the problem statement, review confidential use cases, identify industry gaps and existing solutions, and establish the objectives and scope of the tool.
- Develop an acceptance and validation report to assess testing outcomes against the defined user requirements.
- Document key engineering assumptions, limitations, and boundary conditions under which the tool outputs are considered valid.
- Define high-level safety, risk, and misuse considerations, including consequences of incorrect fuse or cable selection.
- Define operating conditions and critical system requirements, including circuits that must remain operational during normal use and high-demand scenarios, and establish acceptable trade-offs between electrical protection and system availability.

Phase 2: Asset Database Development (Heavy Machinery and Light Vehicles)

- Compile relevant automotive and wiring standards for fusing and cabling, including applicable compliance requirements.
- Develop a library of common automotive wiring configurations used in fleet and specialist applications.
- Document commonly used makes and models of light vehicles and heavy machinery within fleet applications.
- Record electrical and mechanical specifications for circuits typically unfused by the OEM, as well as common aftermarket additions requiring cabling and fuse calculations (where available).
- Capture environmental and installation conditions (temperature, vibration, routing, bundling, enclosure, engine bay vs chassis).
- Include duty-cycle and load-profile information (continuous, intermittent, cranking, peak loads).
- Incorporate cable derating factors such as ambient temperature, grouping, insulation type, and installation method.
- Define standards traceability within the database, linking assets and configurations to applicable clauses or guidance.

Phase 3: Tool Development with Validation of Standards-Based Formulae in Excel

- Identify and document the standards-based formulae governing cable and fuse selection, including required parameters such as inrush and cranking currents, operating temperature, continuous versus breaking current, cable length, and system voltage etc.
- Develop a user-friendly tool in Excel that integrates standards-based formulae and is driven by the asset database.
- Apply and validate these formulae against real-world case studies (confidential) to assess suitability, identify gaps or unrealistic assumptions, and refine the calculations as required.
- Conduct sense-checks with field technicians and project managers to confirm alignment with testing results and practical field experience.
- Incorporate fault current, short-circuit, and protection coordination considerations, including fuse I^2t and cable thermal withstand checks.
- Implement input validation, error handling, and warnings for incomplete, incorrect, or out-of-range inputs.
- Define quantitative acceptance criteria and validation metrics for calculation results.
- Perform edge-case and worst-case scenario testing, including extreme temperatures, maximum fault currents, and long cable runs.
- Document all calculation logic, assumptions, and conservative margins within the tool for transparency and auditability.

Phase 4: Creation of a mobile based- web application using Suitable Software language (e.g. Python, MATLAB).

- Create a defined functional list, required data representation, aesthetic requirements for the tool.
- Update the tool to drive results from the user input that references databases in the background.
- Utilise the tool against phase 3 results to identify any gaps in data representation, software bugs, limitations etc.
- Deploy the tool for verification and validation with head technician and other engineers.
- Record feedback and create a plan to update the tool.
- Create a troubleshooting & process guide to repair any issues with the control logic and for any future development.
- Separate user interface, calculation logic, and databases to improve maintainability and scalability.
- Implement version control and change-tracking for formulas, databases, and standards.
- Develop user guidance and worked examples to support correct interpretation of results.